

Floods and food security: A method to estimate the effect of inundation on crops availability



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ARTICLE INFO

Article history:

Received 19 November 2016

Revised 31 May 2017

Accepted 22 June 2017

Available online 28 June 2017

Keywords:

Water Footprint

Human Energy Requirements

Bangladesh 2007

Pakistan 2010

Remote sensing data

ABSTRACT

The inner connections between floods and food security are extremely relevant, especially in developing countries where food availability can be highly jeopardized by extreme events that damage the primary access to food, i.e. agriculture. A method for the evaluation of the effects of floods on food supply, consisting of the integration of remote sensing data, agricultural statistics and water footprint databases, is proposed and applied to two different case studies. Based on the existing literature related to extreme floods, the events in Bangladesh (2007) and in Pakistan (2010) have been selected as exemplary case studies. Results show that the use of remote sensing data combined with other sources of onsite information is particularly useful to assess the effects of flood events on food availability. The damages caused by floods on agricultural areas are estimated in terms of crop losses and then converted into lost calories and water footprint as complementary indicators. Method results are fully repeatable; whereas, for remote sensed data the sources of data are valid worldwide and the data regarding land use and crops characteristics are strongly site specific, which need to be carefully evaluated. A sensitivity analysis has been carried out for the water depth critical on the crops in Bangladesh, varying the assumed level by $\pm 20\%$. The results show a difference in the energy content losses estimation of 12% underlying the importance of an accurate data choice.

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1. Introduction

Water and food are two key drivers of the global system sustainability. The understanding of their complex interrelations, the water–energy–food nexus (SEI, 2011; Vanham, 2016; Rulli et al., 2016), represents an important step to enhance the resilience of the water–food systems; as well as to achieve an appropriate management of the natural resources. Therefore, it is evident that food security is strictly related to water and both are highly vulnerable to continuously changing climatic patterns (Misra, 2014; Davis et al., 2017; Chiarelli et al., 2016). Hydrological extreme events are already causing major water crisis within the global systems, jeopardizing food security. The climate change report (IPCC, 2014) shows an increasing variability in rainfall patterns, as well as demonstrating evidence of flood events increasing in many regions of the planet, according to the Millennium Ecosystem Assessment (MEA, 2005). In addition, land use change can enhance flood risk especially in flash flood prone areas (Rosso and Rulli, 2002). For the above reasons, it has become important quantifying the effects of

extreme hydrological events on food production and consequently to food security (Lesk et al., 2016).

Developing countries are mainly affected by these events due to the lack of necessary and essential resources, infrastructures, and adequate disaster management systems (D'Odorico and Rulli, 2013). Furthermore, the effects of natural disasters in developing areas tend to be aggravated by the lack of economic resources, the frequent political instability, wars and conflicts (Zulqarnain, 2013).

According to FAO (2015), between 2003 and 2013, natural hazards and disasters in developing countries have affected more than 1.9 billion people and caused over USD 494 billion in estimated damage. The crop production is the most affected subsector (total damages amount to about USD 13 billion) and floods are the most impacting natural disasters on agriculture, with a percentage share of damage around 60% (FAO, 2015). Food production will have to increase by 70% in order to feed a population exceeding to nine billion people by 2050 (FAO, 2009). Indeed, it is fundamental to manage agriculture systems, in order to preserve their ecosystem services; such as their capacity to assure food security. According to the definition elaborated by FAO (2001b), food security “exists when all of the people, at all times, have physical, social and economic access to sufficient, safe and nutritious food, in

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order to meet their dietary needs and food preferences for an active and healthy life". In 2009 Food and Agriculture Organization of the United Nations identified the four pillars of food security as availability, access, utilization, and stability (FAO, 2009).

Flood risk is connected to all the four dimensions of food security. In fact, food availability looks at the "quantities of food available on a consistent basis" and the expression "food access" refers to the possibility to have access to enough food for a healthy diet. On the other hand, food use is defined as the "appropriate use based on knowledge of basic nutrition and care, as well as adequate water and sanitation", while food stability refers to "the temporal determinant of food security and affects all three physical elements" (FAO, 2001b). Hereby, we focus on the effects of flood on food availability, in order to develop a methodology to estimate the effects on the short term food security, i.e. food availability. In this study, the evaluation of food security losses is based upon the integration of remote sensing data and national statistics, which are used to estimate the crops damaged or destroyed by the flood event. On the existing literature-based related to extreme floods, the floods in Pakistan (2010) and Bangladesh (2007) have been selected as exemplary and complementary case studies, which are based on the effects of flood events that make a negative impact on food availability. The analysis related to Pakistan aimed at a general assessment of the effects of flood that affects various types of crop, as well as food stocks. While the Bangladesh case study focuses on rice as the most spread crop found in the country, in order to underline the importance of considering the peculiarities of each crop in terms of flood resistance. The results are then compared to the average annual food supply and the associated energy content to estimate the direct impacts on food security as lost calories. Additionally, the crops losses are reported in terms of Water Footprint (hereafter WF) (Hoekstra et al., 2011), as a complementary indicator to evaluate from a water management perspective the indirect effects of flood at the national nourishment level in terms of water lost.

2. Materials and method

2.1. Studied areas

The role of agriculture is often studied as a driver of land use change, with potential negative or positive effects on hydrological regimes (Kenyon et al., 2008). On the other hand, hydrological extreme events can cause dramatic consequences on the agricultural land that need to be quantified.

This study aims to assess the effects of extreme flood events on the food security of hazard prone countries. Many databases of extreme natural events have been created in the last decades. The majority belongs to reinsurance companies, which have made an estimation of damages and victims of several events in order to classify their characteristics and specify a threshold for the definition of a natural disaster. In 2002, CRED (the Centre for Research on the Epidemiology of Disasters) has reviewed the existing literature and databases based on natural disasters. As a result, the review showed that the only freely available database on natural disaster is EM-DATA (2015) developed by CRED itself (Guha-Sapir et al., 2015).

On the base of EM-DATA database, Bangladesh and Pakistan have been chosen as exemplar case studies, because of their vulnerability to floods (Mirza, 2002; Sardar et al., 2008) and the importance of agriculture in their territories (FAO, 2016).

These case studies are characterized by significant differences in their agricultural production and allow us to evaluate the effects of flooding on crops with different resistance to submergence (Jaiphong et al., 2016; Butzen, 2016).

Bangladesh is a developing country in Southern Asia, with a population of 158 million people and a land area of 144,000 km², which makes it one of the most densely populated countries with 1100 inhabitants/km². Most of its territory is flat, with elevation ranging between 0 and 90 m a.s.l., and 7% of the country is occupied by water bodies, principally by the rivers Brahmaputra, Padma (the Ganges), Meghna and their tributaries. Its tropical climate is characterized by 4 seasons: pre-monsoon, monsoon, post-monsoon and dry season. Around 80% of the annual rain comes during the monsoon season (from June to October), when floods occur regularly, while cyclones occur during the pre and post-monsoon. During the monsoon season a large portion of the flood water from precipitation and river flow infiltrates into aquifers and it is later used during the dry season.

Around 70% of the population lives in rural areas and 20% is economically active in the agriculture sector, generating around 20% of the country's GDP. 70% of Bangladesh territory is used for agricultural production (FAO, 2016) and the main crop in terms of cultivated area, production and calories intake is rice. According to the Bangladesh Bureau of Statistics (BBS, 2005; BBS, 2009), rice production occurs on more than 80% of agricultural lands and it is grown in three growing seasons (AMAN, BORO, and AUS) that also give the name to the three varieties of crops cultivated.

AMAN grows during the monsoon season when rainfall is plentiful. BORO is restricted to irrigated areas and it grows during the dry season, once the floods have receded. AUS production takes advantage of rainfall during the spring transition toward the monsoon, which enables a short growing season, although the sufficiency of rains varies from year to year (Ruane et al., 2013). BORO is currently the most productive season, followed by AMAN, with AUS considerably lower, mostly due to smaller planted area (FAO, 2008), Fig. 1.

Irrigation during the monsoon and post-monsoon seasons is secured by the abundant rain, while groundwater is used for the remaining of the year. Therefore, this means that the agricultural sector, and consequently the country's ability to feed itself, relies heavily on rainfall and the capacity of its management. According to FAO FBS-Food Balance Sheets (FAO, 2016) Bangladeshi food supply in 2013 was 2450 kcal/day (2344 and 106 from vegetal and animal products respectively and mainly contributed (76%) by cereals, in particular rice (92%). Domestic production covers the 88% of total food supply, being rice almost exclusively produced in Bangladesh. Food import is 12%, being vegetal oil, pulses and sugar the major imported items and food export is 0.5%. FAO food security indicators report the 3-year average (2013–2015) depth of the food deficit equal to 120 kcal/cap/day and prevalence of undernourishment and of food inadequacy equal to 16.9% and 26.6%, respectively (FAO, 2016).

Floods are usual components of Bangladesh ecology, but extreme events (such as the ones of 1998 or 2007) can severely affect the national agricultural production, threatening the food security of tens of millions of households (Del Ninno et al., 2003).

The geographical setting of Bangladesh contributes to a historically high number of extreme floods: six extreme flood events (recurrence interval > 100 years) have occurred since the year 1985 (Kale, 2014), mostly caused by overabundant rainfall and inundation of larger rivers, causing considerable damages to all sectors, but principally to the agricultural sector.

In 2007, between early June and September, massive floods caused by the monsoon rains hit the south of Asia. Bangladesh, among the neighboring countries, was severely damaged. The floods involved the Padma and Brahmaputra river basins, affecting the nine districts of Dhaka, Munshiganj, Sunamganj, Rajbari, Madaripur, Shariatpur, Manikganj, Netrakona, Jamalpur, Kishoreganj and Tangail.

Pakistan is a developing country of South Asia, with a population exceeding 191 million people it is among the world most



Fig. 1. Bangladesh crop calendar (adapted from FAO, 2008).

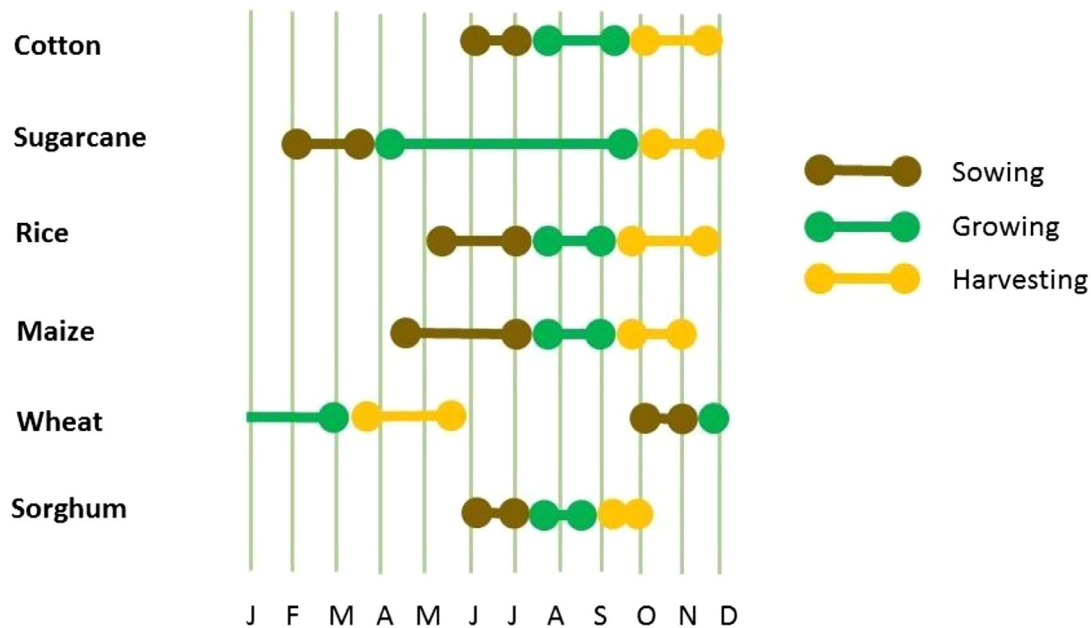


Fig. 2. Pakistan crop calendar (adapted FAO, 2012).

populated country. Pakistan covers an area of 796,095 km² and it is divided into three major geographic areas: the northern highlands, the Indus River plain (that includes Punjab and Sindh) and the Balochistan Plateau.

The climate varies from tropical to temperate. Arid conditions exist in the coastal south, characterized by a monsoon season with adequate rainfall and a dry season with lesser rainfall, while abundant rainfall is experienced by the province of Punjab. Half of the annual rainfall occurs in July and August.

Pakistan is a rapidly developing country. Fifty percent of its territory is used for agriculture and this sector accounts for 25.3% of GDP and employs about 43% of the labor force. Punjab province has the highest share of agricultural production, mainly wheat and cotton followed by rice (FAO, 2016) (Fig. 2).

Pakistan's food production depends greatly on irrigation, which provides more than 90% of the country's wheat and most of the other crops (FAO, 2016).

According to FAO FBS (FAO, 2016) Pakistan food supply in 2013 was 2440 kcal/cap/day (1910 and 530 from vegetal and animal products respectively and mainly contributed by cereals (48%), in particular wheat (903 kcal/cap/day). Domestic production covers almost the total food supply in terms of wheat and around 75% of rice.

Food import is very low (3%), being palm oil the major imported item and food export is 5% (mainly rice, sugar and wheat). FAO food security indicators report 3-year average (2014–2016)

depth of the food deficit equal to 172 kcal/cap/day and prevalence of undernourishment as well as of food inadequacy, which is equal to 22% and 30.5%, respectively (FAO, 2016).

Statistics show that Pakistan is a country prone to flood: 21 floods occurred between 1950 and 2010 in the Indus Basin, causing cumulative direct economic losses of about \$19 billion, killing 8887 people, and damaging or destroying a total of 109,822 villages (Ali, 2013). The 2010 Pakistan flood hereby analyzed has been the most damaging event on record, causing around 1600 fatalities. The flood involved the Indus river basin, affecting especially Khyber Pakhtunkhwa, Punjab and Sindh.

2.2. Method

The effects of floods on agriculture and the related losses in terms of food security can be calculated integrating remote sensing data with land use assessment and agricultural statistics.

Firstly, the evaluation of potentially affected agricultural land is given by the intersection of remote sensed flood maps with administrative and land use maps of the area of interest. The classification of the flooded area can be done following one of the many existing algorithms on the base of raw available data or using existing remote sensing product, already classified for flood identification (Sanyal and Lu, 2004).

The potential agricultural land losses are later converted into effective crop loss taking into consideration the peculiarities of topography, land use and crops characteristics for the selected study

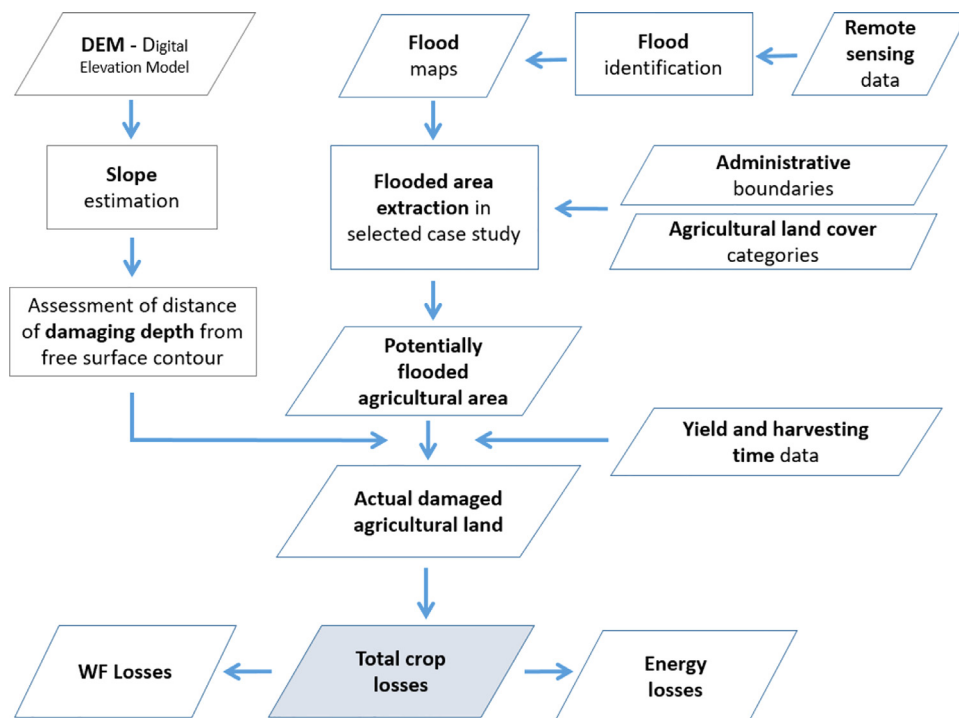


Fig. 3. Crop losses estimation methodology scheme.

area. Additional information, such as specific crop resistance to submergence in terms of time and water depth might be added to refine the crop losses evaluation (Fig. 3).

The effects of flood on food availability, thus on food security, is then evaluated both converting crop losses into lost calories and Water Footprint (WF).

Lost calories can be estimated using the specific crop energy content (Schakel et al., 1997) and compared to reference Human Energy Requirements (hereafter HER) (FAO, 2001a). HER is defined as “the amount of food energy needed to balance energy expenditure in order to maintain body size, body composition and a level of necessary and desirable physical activity consistent with long-term good health”. The HER varies vastly in terms of age, sex, body weight, physical activity and general lifestyle. According to FAO (FAO, 2001a) a mean Human Energy Requirement is here approximated at a value of 3000 kcal/cap/day. Furthermore, a minimum Human Energy Requirement is approximated at 1800 kcal/cap/day. Taking into account the population, a per capita loss value is determined and compared to food availability levels prior to the floods, in terms of the total energy lost per capita and HER.

On the other hand, the WF associated to flood events can be evaluated from the lost crops, using the crop’s WF values available in the existing literature (Mekonnen and Hoekstra, 2010). The results can be then compared to the total WF of an averaged country diet and food supply, derived from FAO data (FAO, 2016).

The WF is an indicator that looks at the direct and indirect use of water of a product. It consists of three components: the green water footprint, the blue water footprint and the gray water footprint. The green WF is the use of rainwater, the blue WF is the use of surface and underground water, while the gray WF refers to the amount of water needed to dilute the water that has been polluted, considering local water quality standards.

In this study, the assessment of WF associated to food losses represents an additional assessment of the vulnerability of a food production system in terms of water availability. In fact, food losses represent not only a lost in calories that directly support human

wellbeing but also a loss of water (particularly the blue water embedded into crops, i.e. irrigation) that can influence the future food production.

The use of the two metrics (i.e. energy content and WF) allows identifying the criticalities of land use management both for supporting a complete diet and preserving local water resources.

If the percentage of losses with respect to the total country food supply in term of calories is higher than WF, this might indicate the necessity of improving the cultivation management (e.g. changing the crop distribution in flood prone areas) to reduce the risk of losing high calorie crops, fundamental to support local diet especially in food insecure countries not connected to the food market.

By contrast, if the percentage of losses in term of WF is higher than in term of calories it could mean that it is necessary to improve crop and water management to reduce, especially in high water scarcity areas, potential blue water losses.

3. Application and results

3.1. Bangladesh: single crop losses

According to the methodology described in Section 2.2, the potential affected agricultural land can be estimated making use of remote sensed maps of 2007 flood with administrative and land use maps of Bangladesh.

In this study the remote sensing data provided by the Global Hazards Information Network and UNOSAT are used. In particular, UNOSAT provides the flood map of the 2007 event, based on MODIS Aqua satellite data. This data set is the flood analysis for Bangladesh, which began on 2nd August 2007. It includes 30,233 satellite detected water bodies with a spatial extent of 72,972 km² derived from the MODIS-Aqua image acquired on August 2007 and analyzed by UNITAR-UNOSAT (2007) (Fig. 4). The land use data are distributed by the Pacific Disaster Center, as institution reference of the Global Hazards Information Network (GHIN, 2008).

Only rice crops are considered, because rice provides almost all of Bangladesh’s alimentary needs (BBS, 2009). It is assumed that

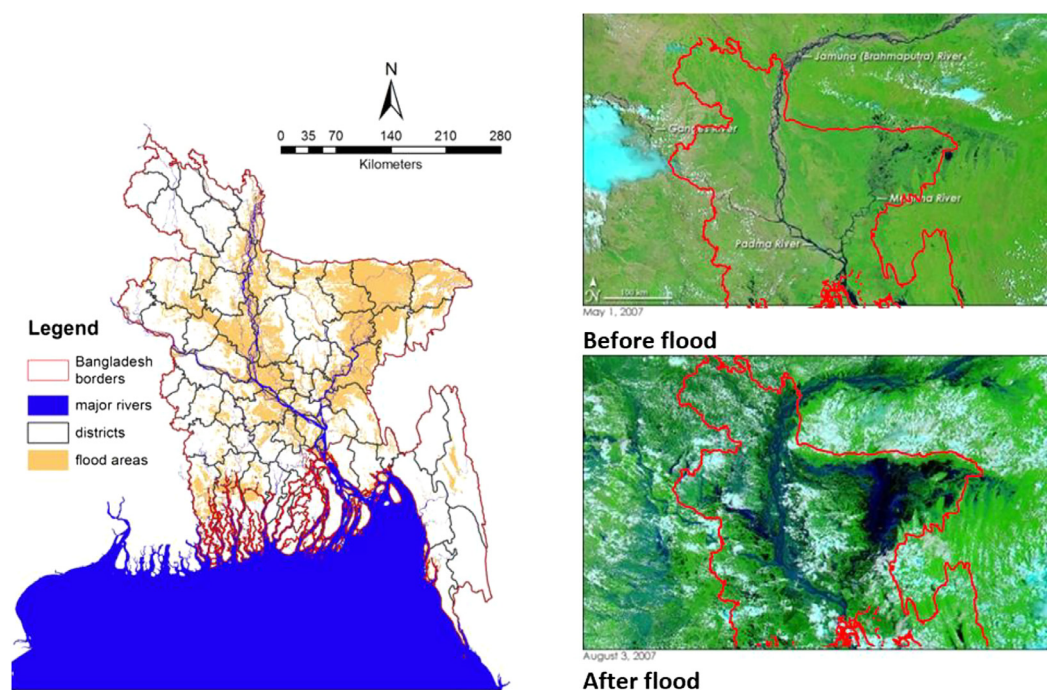


Fig. 4. Bangladesh 2007 flooded area, as identified by MODIS Aqua satellite data.

Table 1
Preliminary assessment of crop losses estimated by remote sensing data.

	Total agricultural area [10 ⁵ ha]	Total agricultural area lost [10 ⁵ ha]	Yield [ton/ha] (BBS, 2009)	Potential production in 2007 [10 ³ ton]	Production losses [10 ³ ton]
AUS	9.9	0.9	1.66	1648.7	149.7
BORO	39.3	16.8	3.86	15,142.4	Not cultivated in 2007 flood
AMAN	60.8	13.2	1.91	11,612.5	2525
Total				28,403.6	2674.7

all of the agricultural areas follow the FAO crops calendar scheme reported in Fig. 1, so that only AMAN and AUS rice are considered. In fact, when the floods occurred, the harvesting of the 2007 BORO paddy crop, accounting for some 30% of the annual production, had been virtually completed, while the AUS crop was being harvested and the main AMAN crop was being planted.

Based on yield values derived from the database of the Bangladesh Bureau of Statistics (Alam and Islam, 2013; BBS, 2009) and the remote sensing data of the flooded cultivated area, the hypothetical losses can be calculated, assuming that the flooded agricultural areas lost all the cultivated crops. Similarly, the potential production of rice for 2007 is hereby calculated, considering that all the agricultural area is productive (Table 1).

The use of remote sensing data for the estimation of crops production lost can be undermined by the uncertainties related to land use and flood extent estimation. A preliminary check can be carried out by comparing the potential production estimated by satellite data to the actual production (provided by the official sources as FAO and Bangladesh National Bureau of Statistics). In this way it is possible to quantify the uncertainty in the estimation of the potential rice production. As shown in Table 2, the estimation of potential rice production made by the use of remote sensing data has a difference less than 5% that is considered acceptable.

On the other hand, the total of lost crops derived by the intersection of land use and flooded areas remote sensed data (i.e. 2674.7×10^3 tons, equals to 10% of the potential rice production for 2007), which results in an unrealistic number compared to

Table 2
Evaluation of remote sensing data performances in crop production estimation.

	Total production [10 ³ ton]	Difference [%]
Remote sensing data	28,404	—
BBS data (BBS, 2009)	28,931	1.8
FAO data (FAO, 2016)	27,196	−4.4

the other existing lost crop estimation made after 2007 flood. (i.e. 1200×10^3 tons, Habiba et al., 2015). This is due to the assumption that in the flooded area identified by the remote sensing all the crops are destroyed. On the contrary, floods affect rice production proportionally to flood hydraulic height and length of the submergence period.

Therefore, a threshold in time and space has been adopted to take into account the resistance of submergence of rice: 1 m depth (according to deep water areas definition made by Huke and Huke, 1997) and a period of submergence greater than 3 weeks (according to the International Rice Research Institute, 1976) have been assumed, as the limiting factors to rice growth.

Therefore, MODIS satellite images of the 20th August were considered and the flooded area that has a water depth below 1 m was excluded from the analysis on the base of the topography SRTM data (van Zyl, 2001).

In order to calculate the area included within the 0–1 m depth, the slope of the DEM (SRTM) is calculated and used to estimate the average distance from the points where the water depth is 1 m

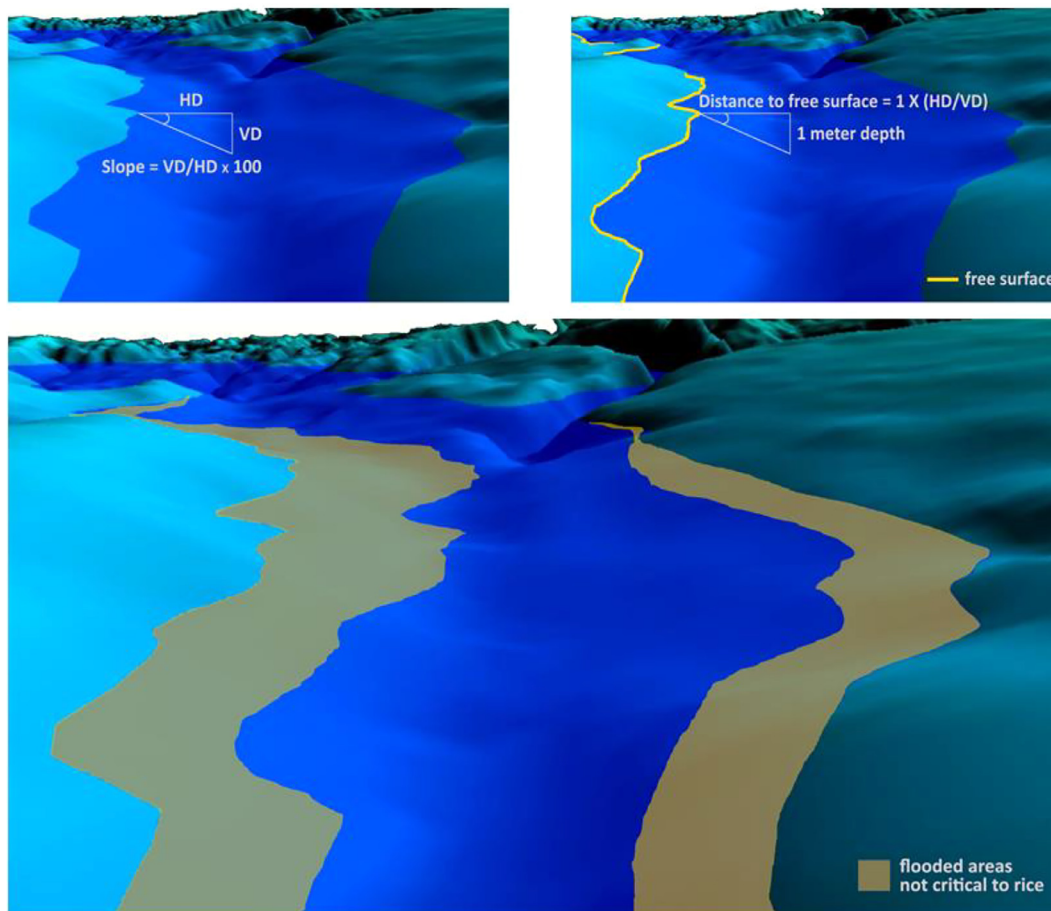


Fig. 5. Estimation method of flooded areas not critical to rice for an assumed threshold of 1 m water depth.

Table 3
Percentages of production lost.

	Potential Production of 2007 [10^3 ton]	Actual Production losses [10^3 ton]	% lost	Net production [10^3 ton]
AUS	1648.7	84.5	5.1	1564.2
AMAN	11,612.5	1571	13.5	10,041.5
Total	13,261.2	1655.5	12.5	11,605.7

to the 0-flood water level (i.e. the flood area perimeter line). The cultivated area not destroyed by the flood is therefore the product of the flood perimeter, times the average distance introduced before (Fig. 5).

The use of MODIS images from the 20th August (6 weeks after the beginning of the flood), guarantees that the time constraint is included.

Results show that 5373 km² of the crop flooded area is between 0 and 1 m of water depth and thus it is not destroyed. Therefore, this means that the actual rice losses decrease to 1655.5×10^3 tons and the estimated agricultural area lost is 8728 km² in accordance with existing estimation, respectively 1200×10^3 tons (Habiba et al., 2015) and 8900 km² (BDER, 2007).

The percentage of lost crops on the potential agricultural production is around 12.5% of the total potential production: this underlines the heavy effect of flood on food production (Table 3).

Once the actual production losses are calculated it is then possible to estimate the effects on food security by converting the losses to an energy value, i.e. kcal.

By knowing the energy content in kcal/kg of rice crops, which in 2007 was an average of 3693 kcal/kg (Wrigley et al., 2015), the quantity of rice lost crops is expressible in calorie supply not provided. The results of Table 4 show the food energy losses considering that only 90% of the total production of rice is destined for human use (percentage obtained from Food Balance Sheets, FAO 2016).

The findings of Table 4 provide the whole framework of the effects of flood on daily availability of food for Bangladesh population in 2007 (146,592,687; World Bank, 2016): 103 kcal lost represent 5.3% of the potential energy provided by the rice without flood and it is almost equal to the local food deficit. If we consider the effects of flood only on the population of the affected districts (i.e. 37,908,436), the result increases dramatically to 398 kcal lost.

The energy deficit due to the rice production lost, compared to the annual energy provided by the entire food production of Bangladesh in 2006 (FAO, 2016), using the total kcal/cap/day supply of 2006 (i.e. 2417 kcal/cap/day) is 4.3% if we consider the total population as affected and it raises to 16.5%, if only the population of the affected areas is considered.

To properly understand the meaning of these percentages, it is important to underline that the lack of food (and thus energy) caused by the flood is contributing to worsen the already critical situation of food supply in Bangladesh. In fact, the energy available from food in Bangladesh is already below the HER, as defined by FAO (3000 kcal/cap/day); comparing the daily per capita calories available in non-flood and post-flood scenario to HER it is possible to have a clearer framework of the food security situation in the country. Bangladesh was already suffering a 19.4% deficit in achiev-

Table 4
Estimation of food energy losses.

	Production losses for human use [10 ³ ton]	2007 flood energy lost [kcal]	per capita energy lost [kcal/cap/day] [*]	per capita energy lost [kcal/cap/day] ^{**}
AUS	76.1	2.81E+11	5	20
AMAN	1413.9	5.22E+12	98	377
Total	1490	5.50E+12	103	398

^{*} Evaluated considering the total population as affected.

^{**} Considering only the population of the affected areas (i.e. 37, 908,436).

Table 5
Unfulfilled Human Energy Requirements—HER.

		Difference from HER	% of HER unfulfilled
Total food kcal available (production 2006)	2417	583	19.4
2007 estimated per capita kcal available	2314	686	22.9
2007 estimated per capita kcal available (population of the flooded areas only)	2019	981	32.7

ing HER standard and after the flood the percentage of unfulfilled HER has raised up to 22.9% or 32.7%, if we consider only the population of the flooded areas (Table 5).

As presented in Section 2.2, the last step was to translate the obtained results in terms of water footprint and to estimate the number of people affected by the flood.

The amount of water lost with the loss of crops can be calculated using the existing literature data for agricultural products in Bangladesh (Mekonnen and Hoekstra, 2010), containing country's green, blue and gray water footprint data for many types of agricultural products. In particular, for rice crops, these are the reference values: 2118 m³/ton of green WF, 309 m³/ton of blue WF, 427 m³/ton of gray WF. Using the previously obtained data for the quantity of agricultural goods lost, and the provided water footprint of rice for Bangladesh, a volumetric value of the amount of water wasted is obtained (Table 6).

Considering the WF associated to the average food consumption in Bangladesh, which is equal to 736.9 m³/year/cap (Hoekstra and Mekonnen, 2012), it is therefore possible to check the estimation made using energy content as a reference for food security (Table 7).

This estimate represents the attempt of translating the WF analysis of the crops lost due to flood into an estimation of flood magnitude. The value obtained is very similar to the value obtained considering the energy losses (i.e. 4.3%), this is due to the fact that the Bangladeshi diet is based on rice. The WF is used as an indicator to evaluate the effects of a flood event to the national nourishment level, representing an alternative and complementary method that can be effective in highlighting the hotspots in a country flood management.

3.2. Pakistan: multi crops losses

According to the methodology described in Section 2.2, the affected agricultural land is estimated making use of remote sensing derived data that are provided by Suparco and FAO (2010), based on MODIS Aqua satellite and SPOT VGT data. The flood involved an area with an extension of 58,797 km² with duration of around 3 months, from the end of July until the end of October (Fig. 6).

All the main Pakistan crops have been considered with detailed data for cotton, sugarcane and rice (Suparco and FAO, 2010). It is

assumed that all the agricultural areas follow the FAO crop calendar scheme reported in Fig. 2. Therefore, in Punjab and Sindh, cotton and other crops such as sugarcane were directly destroyed by the flood, while wheat has been damaged as stock.

The characteristics of the crops in terms of yield can be derived from FAOSTAT database (FAO, 2016).

Based on the data provided by Suparco and FAO (2010) the cultivated areas that were flooded are estimated. Results show a total of 23,640 km² of agricultural areas destroyed, with Punjab and Sindh being the districts more affected by the flood (Table 8).

Using the FAOSTAT, the crops production data for the 10 years before the flood event (FAO, 2016), it is possible to calculate an average agricultural yield for the main crops lost (i.e. sugarcane and rice), as well as to estimate the actual crop losses (in term of weight). By knowing the energy content in kcal/kg of the different crops, it is then possible to estimate the associated food energy losses, both in terms of vegetal and animal calories (Rulli and D'Odorico, 2014; Davis et al., 2014) (Table 9).

Results show a reduction of production that is around 19% for sugarcane and 40% for rice (comparison made with 2009 production data (FAO, 2016)) that is associated to a significant loss of energy available.

Beside the losses of crops directly destroyed in the fields, the food stocks have been also damaged. Based on literature data (Impact Forecasting, 2010) also the damages to wheat stocks have been considered (Table 10).

The sum of crops and stocks destroyed amounts to a total of 205 kcal/cap/day lost, due the flood and it is equal to a loss of 8.5% of the Pakistan average food supply (10.7% if we consider only the energy derived from vegetal products).

As shown in the Materials and method section, Pakistan food supply was already below the HER requirements with a gap of 571 kcal/cap/day in 2009 (FAO, 2016). The effects of flood caused an average increase of HER unfulfilled around 26%. The results are even worse if we subdivide the result for the different provinces (Balochistan unfulfilled HER rises to 31%), in particular for Sindh province, in which the values move close to the minimum HER of 1800 kcal/cap/day (Table 11).

Based on the existing statistics on WF in Pakistan (Mekonnen and Hoekstra, 2010) the lost food is also translated in terms of water footprint in order to have another measurement of the flood effects on the territory. Results show a total WF of 1.84E+10 m³ that is equal to 13.5% of the Nation WF of the country (Table 12).

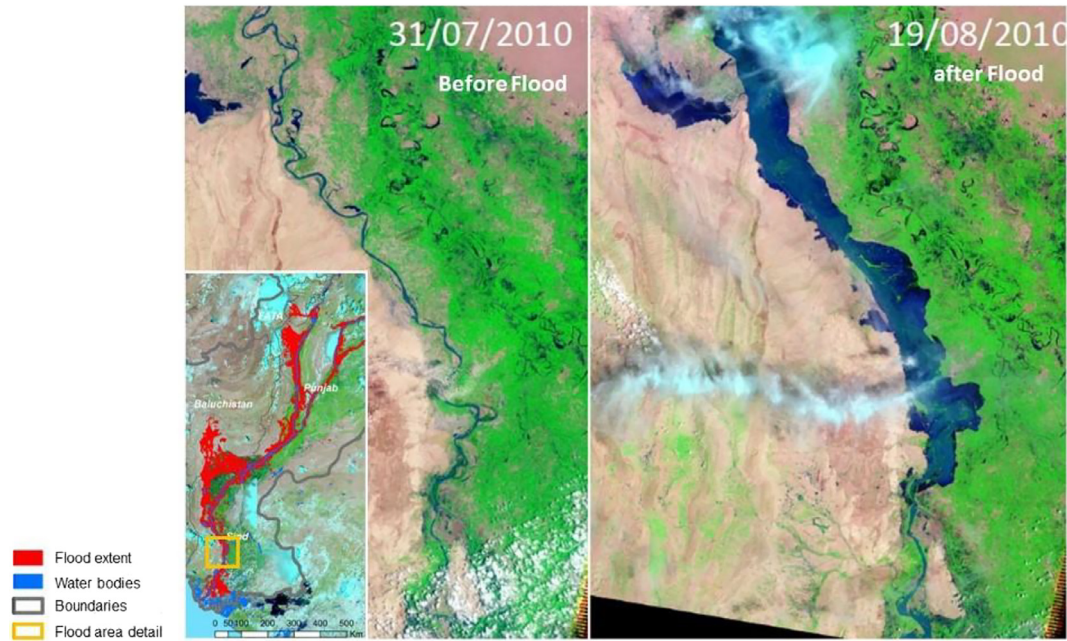
4. Discussion

The method proposed is based on the integration of different sources of information, such as remote sensing data, agricultural statistics and water footprint databases.

The application has demonstrated the capability and the criticalities of the method in determining the consequences of floods to food supply. It has specifically been shown that the remote sensing data by itself lacks the level of accuracy necessary in a comprehensive food security assessment. On the other hand, the use of remote sensing data, if sustained by other onsite information, can

Table 6
Water Footprint—WF rice lost.

Rice type	Green WF [m ³]	Blue WF [m ³]	Gray WF [m ³]	Total WF [m ³]
AUS	1.79E + 08	2.61E + 07	3.61E + 07	2.41E + 08
AMAN	3.33E + 09	4.85E + 08	6.71E + 08	4.48E + 09
Total	3.51E + 09	5.12E + 08	7.07E + 08	4.72E + 09

**Fig. 6.** Pakistan 2010 flooded area, as identified by MODIS Aqua satellite data.**Table 7**
Affected population estimated by Water Footprint—WF analysis.

Total lost water	4.72E + 09
Consumption per capita	7.37E + 02
National WF	1.08E + 11
% of total WF lost	4.4%

be particularly useful to assess the effects of flood on an agricultural area.

This analysis framework is general and replicable to any location for what concern the use of remote sensed data; some limitation may arise for the data regarding land use and crops characteristics which are strongly site specific and often not easy to collect. A sensitivity analysis has been carried out for the water depth critical to crops in Bangladesh, varying the assumed level by $\pm 20\%$. The results show a difference in the energy content losses estimation of 12% underlying the importance of an accurate choice.

This framework of analysis can help locating the hotspots as high flood risk areas where the crops provide high level of energy or have high water footprint. These are the areas that require proper protection to assure the necessary food production and the

Table 9
Estimation of food losses.

District	Crop Losses	
	Sugarcane	Rice
Total lost area [10 ⁵ ha]	1.95	8.73
yield [ton/ha]	48.81	3.15
Total lost production [10⁶ ton]	9.5	2.75
Energy content [kcal/kg]	300	2800
Energy losses _{veg} [kcal]	2.85E + 12	7.7E + 12
Energy losses _{veg} [kcal/cap]	45.8	123.7

Table 10
Estimation of wheat losses.

Districts	Wheat damaged in stocks [10 ³ ton]
Balochistan	0
Khyber PK	80.8
Punjab	44.9
Sindh	541.7
Total	667.4
Energy content [kcal/kg]	3340
Energy losses_{veg} [kcal]	2.23E + 12
Energy losses_{veg} [kcal/cap]	35.8

Table 8
Districts flooded areas assessment.

Province	Cotton [10 ³ km ²]	Sugarcane [10 ³ km ²]	Rice [10 ³ km ²]	Other crops [10 ³ km ²]	Total [10 ³ km ²]
Balochistan	0.03	0	1.24	0.20	1.47
Khyber PK	0	0.15	0.06	0.41	0.62
Punjab	4.05	1.03	2.36	4.96	12.40
Sindh	1.91	0.76	5.07	1.37	9.15
Total	5.98	1.95	8.73	6.95	23.64

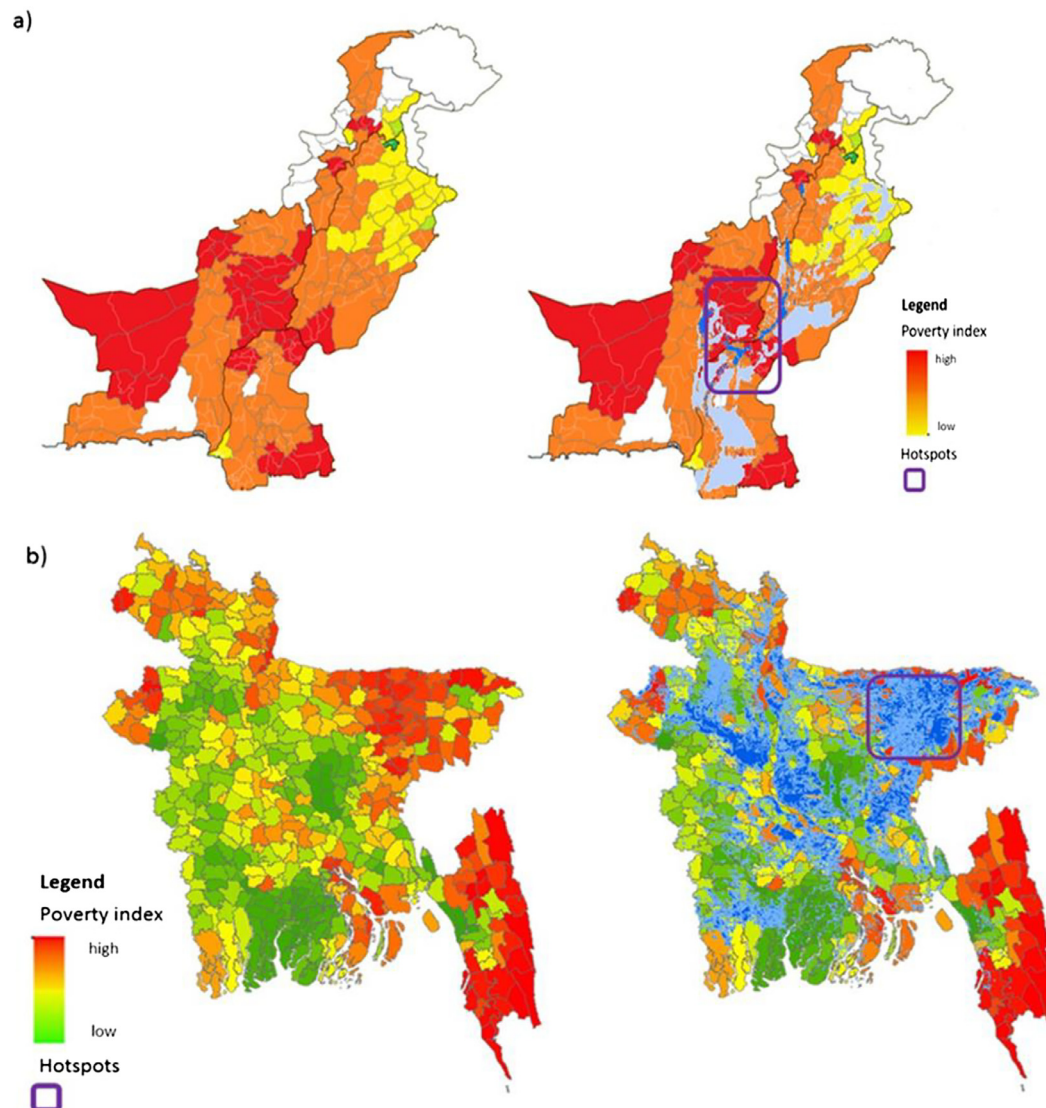


Fig. 7. Identification of the poorest area affected by flood (a) Pakistan (adapted from Arshad, 2005), (b) Bangladesh (adapted from Ahmed et al., 2010).

Table 11
Spatial distribution of unfulfilled HER.

Provinces	Energy losses [kcal]	Province population [cap]	Energy losses per capita [kcal/cap/day]	Per capita energy supply 2010 [kcal/cap/day]	Unfulfilled HER
Balochistan	1.09E + 12	8679,650	346.1	2077	31%
Khyber PK	0.54E + 12	22,567,090	66.0	2357	21%
Punjab	3.74E + 12	93,740,220	109.2	2314	23%
Sindh	7.40E + 12	38,190,460	531.1	1892	37%

Table 12
Water footprint of crop losses.

	Green WF [m ³]	Blue WF [m ³]	Gray WF [m ³]	Total WF [m ³]
Sugarcane WF	8.74E + 08	2.06E + 09	2.66E + 08	4.25E + 09
Rice WF	2.89E + 09	9.45E + 09	1.19E + 09	1.24E + 10
Wheat WF	4.89E + 08	9.13E + 08	2.71E + 08	1.73E + 09
Total lost water				1.84E + 10
Annual national WF				13.66E + 10
% of total WF				13.5

proper management to guarantee food security to preserve the available water resources. The two case studies have highlighted the peculiarities of flood effects on countries with different food production systems. In Bangladesh, where the country is basing its food production on one major cultivation the impact in terms of percentage of calories and WF lost is almost equal. On the other hand, Pakistan show a higher percentage of WF losses compared to energy: this is due to the fact that in Pakistan internal food production is based on the cultivation of several crops, with different water requirements. In this particular case, it results that high demanding crops (e.g. sugar cane, rice) are cultivated in the 2010 flood event affected areas.

Moreover the results can be combined with other spatial analysis in order to provide a broader picture of the flood effects on a territory. An integration of flood extent with population poverty distribution is hereby proposed to identify the hotspots areas where flood strikes the poorest areas (i.e. districts of Netrokona, Kishoreganj and Sumanganj in Bangladesh and Tank, Kashmir, Rajanpur, Ghotki, Thatta in Pakistan), Fig. 7.

The strategies for disaster management and planning against flood risk should battle social disparities and focus on these areas that are often more vulnerable and less resilient to flood (Vojinovic and Abbott, 2012). This kind of analysis can provide useful information that help the definition of the priorities of intervention to support the poorest areas in case of flood events.

It should be noted that food security could be strongly reduced by floods damages on pasture land, livestock, fisheries affecting directly or indirectly the animal calories and protein food intake (Davis et al., 2014).

Moreover, this analysis focuses on direct flood damages to crops or food stocks and, other side effects of flood (e.g. deposition of sediment on fields, erosion of agricultural soil, loss of soil nutrients and microbial and fungal activities) that could be an obstacle for future possibilities of cropping, should be taken in account (Pimentel et al., 1995; Zhang et al., 2017). Other factors influencing future production may be the spread of insects that could be dangerous to cultivation due to the presence of stagnant water (Rosenzweig et al., 2001).

5. Conclusion

The present study analyzes the effects of floods on short terms food availability, and thus security, for two different case studies (Bangladesh 2007 and Pakistan 2010). The adopted method integrates remote sensing data, agricultural statistics, and water footprint values in order to (i) evaluating the potentially affected agricultural areas, (ii) converting the affected areas into crop loss, (iii) estimating the associated calories and water footprint losses.

In Bangladesh, the estimated lost rice is around 12.5% of the total potential production, which implies a 5.3% calories loss with respect to the total potential energy provided by rice and 4.4% of total WF associated to national food supply. In Pakistan, the results show a crops loss of 19% for sugarcane and 40% for rice, with a related calories loss of 8.5% and a WF loss of 13.5%.

The results highlighted the countries vulnerability to flood, being both countries strongly dependent on local agricultural production. The 2007 flood event reflected badly upon Bangladeshi food security, almost doubling the existing food deficit. The same happened in Pakistan where an already scarce food supply has been worsened by the 2010 flood.

These case studies stress the importance of integrating different analysis approaches to carry out an assessment of the meaningful connections between flood and food security and to enhance the resilience of territories. From a nexus perspective, this study allows to clarify how flood management could be crucial also for food production. The analysis of calories and WF losses provides a

comprehensive assessment of the inner relationships between food production and water use; in fact, food losses represent not only a lost in calories that directly support human wellbeing but also a loss of water that can indirectly jeopardize food security.

The methodology, taking advantage of the integration of remote sensing data, and agricultural statistics, is general and replicable to any location providing a rapid assessment of flood effects on food security on the short term. Moreover, if combined with other spatial information (e.g. poverty distribution, as shown in the case studies), it can provide useful information that can be applied to spatially identify the hotspots and support long term planning.

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